CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 6: **Control and Coordination** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a reflex arc, the correct order is: 1

(a) receptor -> sensory neuron -> spinal cord -> motor neuron -> effector
(b) effector -> motor neuron -> spinal cord -> sensory neuron -> receptor
(c) receptor -> spinal cord -> brain -> effector
(d) receptor -> motor neuron -> spinal cord -> sensory neuron -> effector

2. Why are shoots positively phototropic and roots negatively phototropic? 1

(a) roots need light
(b) shoots need gravity
(c) roots have no auxin
(d) auxin affects stem and root cells in opposite ways

3. A person has frequent thirst and urination and sugar in urine. Which hormone is deficient? 1

(a) adrenaline **(b)** thyroxine **(c)** insulin **(d)** growth hormone

4. The connection between sensory and motor neurons inside the spinal cord is made by: 1

(a) dendrite **(b)** relay neuron **(c)** effector **(d)** axon

5. If the tip of a shoot is removed, phototropic bending: 1

(a) occurs more **(b)** does not occur **(c)** increases **(d)** reverses

6. A child of age 12 is very short but healthy. The hormone lacking: 1

(a) adrenaline **(b)** growth hormone **(c)** insulin **(d)** thyroxine only

7. A person touches a hot plate and quickly pulls the hand. The effector is: 1

(a) spinal cord **(b)** muscle **(c)** brain **(d)** skin

8. Which hormone is responsible for the wilting of leaves? 1

(a) auxin **(b)** gibberellin **(c)** cytokinin **(d)** abscisic acid

9. A person has a swelling in the neck and is lethargic. The cause may be a deficiency of: 1

(a) iron **(b)** iodine **(c)** sodium **(d)** calcium

10. The impulse travels in a neuron from: 1

(a) dendrite to axon end **(b)** synapse to axon
(c) axon to dendrite **(d)** cell body to dendrite

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are reflexes important for survival? 2

12. Why is impulse transmission one-directional in a synapse? 2

13. Why do we need hormones in plants for growth? 2

14. Differentiate between reflex action and voluntary action. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A person's cerebrum is damaged. Which functions are affected? 3

16. Describe the role of feedback in the control of thyroxine. 3

17. Why are the effects of hormones slower than nerve impulses? 3

OR

Explain how information is passed between two neurons. 3

18. Explain why a person who has an increased level of adrenaline appears pale and sweating, with heart pounding. 3

OR

How is the growth of tendrils useful? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Trace the path of an impulse when you step on a thorn: describe receptors, neurons, brain, response and why the foot moves before pain is felt. 5

OR

Explain how the blood sugar level is controlled and what happens in diabetes. 5

20. Explain the human brain parts and give two functions of each part, linking to everyday actions. 5

OR

Describe the structure of a neuron and explain how an impulse crosses the synapse. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND